

HW-2

1. $CI = 95\%$, $e = 15$ second, $\sigma = 40$

$$\bar{x} \pm \left[Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = e \Rightarrow e\sqrt{n} = Z_{\alpha/2} \sigma$$

$$\sqrt{n} = \frac{Z_{\alpha/2} \sigma}{e} \Rightarrow n = \left(\frac{Z_{\alpha/2} \sigma}{e} \right)^2$$

$$n = \left(\frac{1.96 \times 40}{15} \right)^2 = 28$$

2. $n = 20$, $\bar{X} = 11.3$, $\sigma = 2.45$, $CI = 95\%$
 $\alpha = 0.05$, $\alpha/2 = 0.025$, $Z_{0.025} = 1.96$

$$\sigma^2 = 6.0025$$

CI for $\mu \Rightarrow \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$11.3 \pm \frac{(1.96)(2.45)}{4.472} \Rightarrow 11.3 \pm 1.074$$

$$\Rightarrow (10.226, 12.374)$$

we had to use $t_{0.025, 19} = 2.093$ since $n < 30$

3. $\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}$

$$19 \times 6.0025 = 114.05$$

$$\chi^2_{0.025, 19} = 32.852$$

$$\chi^2_{0.975, 19} = 8.907$$

$$\Rightarrow (3.47, 12.8)$$

4. $C_1 \Rightarrow 103, 94, 110, 87, 98$ (5)

$C_2 \Rightarrow 97, 82, 123, 92, 175, 88, 118$ (7)

$(\mu_1 - \mu_2)$, 90% CI $\alpha = 0.1$, $\alpha/2 = 0.05$

2 unequal variances

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$\bar{X}_1 = 98.4, \quad n_1 = 5, \quad v_1 = 4, \quad S_1^2 = 76.3$$

$$\bar{X}_2 = 110.714, \quad n_2 = 7, \quad v_2 = 6, \quad S_2^2 = 1035.905$$

$$v = 7.19$$

$$t_{\alpha/2, 7.19} = 1.887$$

$$-12.314 \pm (1.887) \sqrt{15.26 + 147.986}$$

$$-12.314 \pm \underbrace{(1.887)(12.777)}_{24.11} \Rightarrow -12.314 \pm 24.11$$

$$(-36.42, 11.796)$$

$$5. \quad \frac{S_1^2}{S_2^2} \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} F_{\alpha/2}(v_2, v_1)$$

$$\alpha = 0.1 \Rightarrow \alpha/2 = 0.05$$

$$F_{0.05}(4, 6) = 4.5337$$

$$F_{0.05}(6, 4) = 0.1623$$

$$\frac{S_1^2}{S_2^2} = \frac{76.3}{1035.905} = 0.074$$

$$\frac{0.074}{4.5337} < \frac{\sigma_1^2}{\sigma_2^2} < \underbrace{0.074(0.1623)}_{0.012}$$

$$0.016 < \frac{\sigma_1^2}{\sigma_2^2} < 0.452$$

Since the CI does not include 1, we could not assume $\sigma_1^2 = \sigma_2^2$

$$6. \quad S'^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

$$(a) \text{ Bias } (\hat{\theta}) = E[\hat{\theta}] - \theta$$

$$E[S'^2] = E\left[\frac{1}{n} \sum (X_i - \bar{X})^2\right] = \frac{1}{n} E\left[\sum (X_i - \bar{X})^2\right]$$

$$= E\left[\frac{n-1}{(n-1)(n)} \sum (X_i - \bar{X})^2\right] = \frac{n-1}{n} E\left[\frac{1}{n-1} \sum (X_i - \bar{X})^2\right]$$

$$E[s'^2] = \frac{n-1}{n} \sigma^2$$

$$\text{Bias}(s'^2) = \frac{(n-1)\sigma^2}{n} - \sigma^2 = \frac{n\sigma^2 - \sigma^2 - n\sigma^2}{n} = -\frac{\sigma^2}{n}$$

(b) ~~when~~ $\text{Bias}(s'^2) \propto \frac{1}{n}$

$$n \rightarrow \infty, \text{Bias}(s'^2) \rightarrow 0$$

$$\lim_{n \rightarrow \infty} \text{Bias}(s'^2) = \lim_{n \rightarrow \infty} \frac{-\sigma^2}{n}$$

$$\frac{1}{n} \rightarrow 0 \therefore \text{bias}(s'^2) \rightarrow 0 \text{ as } n \text{ tends to } \infty$$

7. $n = 50, \bar{X} = 78.3, \sigma = 5.6$

$$\sqrt{n} = 7.071$$

$$\alpha = 0.05$$

(a) $\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$\alpha/2 = 0.025$$

$$Z_{0.025} = 1.96$$

$$78.3 \pm (1.96)(0.792)$$

$$78.3 \pm 1.552 \Rightarrow (76.748, 79.852)$$

had to use t because normal populeⁿ with unknown σ .

(b) $t_{0.05, 49} = 1.6766$

$$78.3 - (1.6766)(0.792) \Rightarrow 78.3 - 1.328$$

$$\underline{76.972}$$

(c) $t_{0.05, 49} = 1.6766$

$$78.3 + 1.328 \Rightarrow 79.628$$

8. $n = 400$, $X = 28$

$$e = 0.05$$

$$\hat{p} = \frac{28}{400} = 0.07$$

$$CI = 95\%$$

$$\alpha = 0.05, \alpha/2 = 0.025$$

$$(a) \quad \hat{p} \pm Z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$$

$$Z_{0.025} = 1.96$$

$$0.07 \pm (1.96)(0.0127)$$

$$0.07 \pm 0.025 \Rightarrow (0.045, 0.095)$$

$$4.5\%, 9.5\%$$

(b) The manufacturer has claimed of defective being at most 5%. So, it is statistically possible.

9. $n = 800$, $X = 368$

$$CI = 95\%$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$Z_{0.025} \approx 1.96$$

$$0.46 \pm 1.96 \sqrt{\frac{(0.46)(0.54)}{800}}$$

$$0.46 \pm 0.0345$$

$$(0.4255, 0.4945)$$

But manager said 44% to 48%.

Since the manager's interval narrower than our 95% CI, $\therefore$, not justified.

10.
$$e = Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \Rightarrow n = \left(\frac{Z_{\alpha/2} \sigma}{e} \right)^2$$

$$w_A = 0.8$$

$$w_B = 0.6$$

$$W = 2 Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$w_A = \frac{2 Z_{\alpha/2} \frac{\sigma_A}{\sqrt{n_A}}}{\frac{2 Z_{\alpha/2} \frac{\sigma_B}{\sqrt{n_B}} + \frac{2 Z_{\alpha/2} \frac{\sigma_A}{\sqrt{n_A}}}{2}} = \frac{0.8}{1.6} = \frac{1}{2}$$

$$w_B = \frac{2 Z_{\alpha/2} \frac{\sigma_B}{\sqrt{n_B}}}{\frac{2 Z_{\alpha/2} \frac{\sigma_B}{\sqrt{n_B}} + \frac{2 Z_{\alpha/2} \frac{\sigma_A}{\sqrt{n_A}}}{2}} = \frac{1.6}{1.6} = 1$$

$$\frac{\sqrt{n_B}}{\sqrt{n_A}} = \frac{1}{2} \rightarrow \frac{n_B}{n_A} = \frac{1}{4}$$

$$n_A = 4 n_B$$

so student A has larger sample size

(b) $w_A = 0.8$, $w_B = 1.6$

$\therefore$ student A is more precise since has smaller interval.

(c) Student A has roughly around 4 times the dataset of student B.